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HRP-AI: A WEB APPLICATION FOR HYPERTENSION
RISK PREDICTION WITH CALIBRATED
EXPLAINABLE AI

A special problem in partial fulfillment
of the requirements for the degree of
Bachelor of Science in Computer Science

Submitted by:

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1 Introduction

1.1 Background of the Study

Hypertension, often referred to as the "silent killer" due to its asymptomatic nature, is a leading non-communicable disease and a principal contributor to cardiovascular morbidity and premature mortality worldwide. In the Philippine context, the burden remains substantial. Recent survey cycles, specifically the 2018 National Nutrition Survey (NNS), report a prevalence of 19.2% among Filipino adults aged 20 to 59, escalating dramatically to 35.0% among the elderly. Despite a reported disease awareness rate of 67.8%, blood pressure control remains critically poor; only 75% of those aware are on treatment, and of those treated, only 27% have their blood pressure successfully controlled [1]. This gap contributes significantly to heart disease and stroke, which remain the top causes of mortality in the country.

Advances in artificial intelligence (AI) and machine learning (ML) enable scalable risk stratification and targeted public-health screening by leveraging large, multimodal datasets. These techniques have been applied across settings to predict cardiovascular risk, specifically utilizing the National Nutrition Survey (NNS) provided by the DOST-FNRI [2], and successful deployments as accessible web applications have been demonstrated [3, 4].

Two practical concerns limit the clinical utility of many ML models: (1) model explainability, and (2) probability reliability. Explainable AI (XAI) methods such as SHAP and LIME increase transparency and clinician trust [5, 6]. Independently, probability calibration techniques ensure that a model's predicted probabilities reflect observed frequencies, a requirement when predictions are used in decision-making workflows. Recent methodological work integrates these concerns through the Calibrated Explanations framework, which produces uncertainty-aware explanations and Wachter-style counterfactuals optimized for minimal clinical perturbation aligned with calibrated probabilities [7, 8].

1.2 Statement of the Problem

Existing ML systems for hypertension risk prediction, such as those developed by Eldem [9] and Naik et al. [3], often prioritize discrimination metrics like accuracy and AUC. However, these systems possibly yield unreliable probability estimates for clinical application since they have not been calibrated nor even been tested with calibration metrics. Current systems aiming for reliable explainability typically focus on post hoc testing of calibration rather than employing formal post hoc calibration methods to actively correct probability distortion, as seen in recent studies using survey data [4, 10].

This omission means that many explainable AI outputs are derived from uncalibrated scores, which misrepresent model confidence and undermine clinical trustworthiness. The current "standard" approach to XAI is not completely ideal due to its nature; explaining the raw output of a black-box model without ensuring its probabilistic reliability can lead to misleading interpretations [11]. Furthermore, XAI and other ML systems

are not without flaws; an explanation that justifies an overconfident but incorrect prediction provides a false sense of security to the clinician [7]. Specifically, a Machine Learning System that utilizes the Philippine NNS to produce discriminative hypertension risk predictions, actively corrects probability unreliability using formal calibration methods (Platt scaling, Isotonic Regression, Venn-Abers), and generates a calibrated risk score, uncertainty interval, and actionable, Wachter-style counterfactuals, has not been systematically implemented and evaluated.

1.3 Research Questions

This study aims to answer the following research questions:

1. How can machine learning models (specifically k-Nearest Neighbors, Naive Bayes, Logistic Regression, Random Forest, AdaBoost, XGBoost, LightGBM, and CatBoost) be developed and evaluated for predicting hypertension risk using the Clinical and Health, Anthropometry, and Dietary Survey Components of the DOST-FNRI 2015 National Nutrition Survey (NNS) dataset?
2. How can a Two-Stage Rigorous Optimization strategy be implemented to enhance model performance, and how can the Calibrated Explanations framework be integrated to provide reliable probability estimates using Platt Scaling, Isotonic Regression, and Venn-Abers predictors?
3. How can Explainable Artificial Intelligence (specifically SHAP and LIME) and bounded Brent's optimization be integrated to extract transparent clinical risk drivers and prescribe physiologically realistic, Wachter-style lifestyle counterfactuals?
4. How can the predictive model, along with its interpretable feature importance, uncertainty intervals, and actionable, Wachter-style counterfactuals, be effectively presented in a web application prototype designed as a decision-support tool for potential users like clinicians or health-conscious individuals?

1.4 Objectives of the Study

The general objective of this study is to develop a web application for hypertension prediction that utilizes a machine learning model enhanced with Calibrated Explainable AI to predict hypertension risk.

1.4.1 Research Objectives

Specifically, this study aims to:

1. **Data Acquisition:** Acquire and secure access to the microdata of the 2015 National Nutrition Survey (NNS) from the DOST-FNRI, specifically the Clinical and Health, Anthropometry, and Individual Dietary components.

2. **Data Merging:** Systematically merge the three distinct datasets using unique respondent identifiers ('hhnum' and 'member_code') to create a unified, multimodal feature set for each respondent.
3. **Target Variable Engineering:** Construct the binary ground-truth variable 'Hypertension' by applying the clinical threshold of Systolic Blood Pressure ≥ 140 mmHg or Diastolic Blood Pressure ≥ 90 mmHg to the averaged blood pressure readings found in the Clinical dataset.
4. **Data Cleaning:** Perform rigorous data cleaning within the NNS framework by filtering out invalid entries, dropping non-relevant anthropometric groups (infants, children, pregnant/lactating women), and removing administrative variables irrelevant to clinical prediction.
5. **Imputing Outliers:** Identify and handle outliers in numerical features using statistical methods to ensure robust model training without discarding valuable clinical signals.
6. **Imputing Missing Values:** Optimize the K-Nearest Neighbors (KNN) imputation strategy by conducting a masking experiment on 1% of the dataset to determine the optimal number of neighbors (k), standardizing numerical variables prior to imputation to ensuring distance metric validity.
7. **Feature Engineering:** Construct new, clinically relevant features such as Body Mass Index (BMI) and Waist-to-Hip Ratio (WHR) from anthropometric measurements, and categorical lifestyle levels (e.g., Smoking Level, Alcohol Level) from raw survey questionnaire responses.
8. **Feature Selection:** Apply a strict collinearity filter, leakage guards, and correlation analysis to select the most predictive variables and reduce dimensionality, thereby improving model efficiency and reducing noise.
9. **Model Development:** Train eight candidate supervised machine learning algorithms, including k-Nearest Neighbors (kNN), Naive Bayes, Logistic Regression, Random Forest, AdaBoost, XGBoost, LightGBM, and CatBoost, to predict binary hypertension status.
10. **Hyperparameter Optimization:** Optimize the performance of each candidate model using a **Two-Stage Cross-Validation (CV) Optimization strategy**, utilizing a broad **Randomized Search** to identify promising regions of the hyperparameter space, followed by **Local Refinement** to pinpoint the configuration yielding the highest F1-score and AUC.
11. **Class Imbalance Handling:** Evaluate and apply imbalance handling strategies, specifically Baseline, Class Weights, SMOTE, SMOTENC, SMOTE + Class Weights, and SMOTENC + Class Weights, to address the disparity between hypertensive and normotensive classes in the dataset.
12. **Probability Calibration:** Apply and evaluate three post-hoc calibration techniques—Platt Scaling, Isotonic Regression, and Venn-Abers predictors—to ensure the predicted risk scores accurately reflect true empirical probabilities.

13. **Explainability Integration:** Implement Explainable Artificial Intelligence (XAI) modules by integrating SHapley Additive exPlanations (SHAP) and Local Interpretable Model-agnostic Explanations (LIME). This integration will generate both global feature attributions (via SHAP) to identify dataset-wide risk drivers, and local attribution scores (via SHAP and LIME) to quantify the contribution of specific clinical and dietary variables to individual risk predictions.
14. **Counterfactual Generation:** Implement the extraction of Wachter-style counterfactual explanations by minimizing an objective function that balances decision boundary crossing with a variance-normalized proximity penalty. The target probability will be set relative to the selected decision threshold determined during model validation, allowing the system to identify minimal clinically realistic single-feature perturbations required to alter a high-risk prediction.
15. **Evaluation:** Assess the models using discrimination metrics (Accuracy, Recall, Precision, F1-Score, AUC) and calibration metrics (Expected Calibration Error, Log Loss) to select the optimal final model.

1.4.2 System Objectives

Specifically, this study aims to:

1. Design and build a prototype web application with a tiered explanation interface to serve as a decision-support tool.
2. Implement an interface for a user to input their required health data, specifically clinical variables (e.g., age, sex, smoking history), anthropometric variables (e.g., BMI, Waist-to-Hip Ratio), and dietary variables (e.g., daily intake of nutrients and food groups).
3. Integrate the trained and calibrated model to generate a personalized hypertension risk prediction based on the user's input.
4. Present the calibrated risk score, uncertainty interval (Venn Abers predictors), and both global feature importance (SHAP Global) and local patient-specific (SHAP Local and LIME) breakdowns for users or clinicians seeking a detailed technical analysis.

1.5 Significance of the Study

This study is significant to the following:

1. **For the Patient:** The system provides personalized hypertension risk estimates, calibrated probabilities, and interpretable explanations that may guide preventive action and support self-management of their health.

2. **For Doctors and Healthcare Practitioners:** The calibrated outputs, uncertainty intervals, and explanation interface offer transparent insights that may assist in clinical assessment, patient counseling, and early intervention planning, ensuring that decisions are backed by reliable probability estimates.
3. **For Government and Policy Makers:** The prototype demonstrates how survey-based predictive tools may support screening workflows and wellness initiatives using national-level data such as the NNS, potentially informing public health policies regarding hypertension screening programs.
4. **For the Research Community:** The study provides a reference implementation for integrating probability calibration with explanation methods, contributing to future work on interpretable and uncertainty-aware predictive models.

1.6 Scope and Limitations

The scope of this study and its limitations are as follows:

1. This study utilizes only the Clinical and Health, Anthropometry, and Dietary components of the 2015 NNS microdata for model training and evaluation.
2. The study focuses on training multiple machine learning algorithms, specifically k-Nearest Neighbors, Naive Bayes, Logistic Regression, Random Forest, AdaBoost, XGBoost, LightGBM, and CatBoost.
3. The study applies specific imbalance-handling strategies, namely Baseline, Class Weights, Synthetic Minority Over-sampling Technique (SMOTE), SMOTENC, SMOTE + Class Weights, and SMOTENC + Class Weights.
4. The study performs probability calibration using Platt scaling, Isotonic Regression, and Venn–Abers predictors.
5. The system integrates SHapley Additive exPlanations (SHAP) and Local Interpretable Model-agnostic Explanations (LIME) within the Calibrated Explanations framework to produce uncertainty-aware explanations. Specifically, SHAP is utilized for both global (population-level) and local (individual-level) feature attributions, while LIME provides accessible local explanations. These are implemented alongside Wachter-style counterfactuals optimized via bounded Brent’s method.
6. The web application developed is a prototype decision-support tool intended for demonstration and exploratory use; it is not intended to replace professional medical judgment.
7. Model performance may be affected by the limitations of NNS data, such as cross-sectional design, measurement error, and reporting biases, particularly in dietary information, as recall bias is inherent in self-reported 24-hour food recall surveys.
8. Generalizability of the hypertension risk prediction model to populations outside the NNS sampling frame or to clinical environments may require further validation and calibration.

1.7 Assumptions

1. Measurements from the National Nutrition Survey (NNS) are assumed to be of adequate quality for hypertension risk modeling.
2. The selected variables from the clinical, anthropometric, and dietary components are assumed to capture meaningful predictors of hypertension risk.
3. Calibration subsets used for Platt scaling, Isotonic Regression, and Venn–Abers are assumed to contain sufficient samples to avoid severe overfitting and yield stable probability estimates.
4. Users of the web application prototype are assumed to provide inputs within realistic ranges consistent with NNS measurement standards (avoiding outliers), which will be enforced via input validation logic within the application.
5. The average systolic and diastolic blood pressure measurements used in this study are assumed to be resting blood pressure readings, consistent with standard clinical practice guidelines for hypertension diagnosis.

2 Review of Related Literature

This section provides a comprehensive critical review of the existing body of knowledge relevant to the development of an explainable and calibrated machine learning system for hypertension risk prediction. The discussion is thematically organized into five key areas: (1) the epidemiological landscape of hypertension, specifically within the Philippine setting; (2) the evolution and application of machine learning methodologies in cardiovascular risk stratification, including modern optimization strategies; (3) the critical role of Explainable Artificial Intelligence (XAI) in fostering trust in clinical decision support systems, specifically distinguishing between interpretability and explainability; (4) the often-overlooked necessity of probability calibration in risk modeling; and (5) a synthesis of these domains to highlight the specific research gap addressed by this study, emphasizing the transition from arbitrary counterfactuals to mathematically rigorous, Wachter-style prescriptive analytics.

2.1 Hypertension: A Global and National Health Crisis

Hypertension, clinically defined as persistently elevated arterial blood pressure, is universally recognized as a primary driver of cardiovascular morbidity and premature mortality. The World Health Organization (WHO) identifies it as a "silent killer" due to its asymptomatic progression towards severe complications, including stroke, myocardial infarction, heart failure, and chronic kidney disease. The condition often remains undetected until significant organ damage has occurred, necessitating proactive screening measures.

In the context of the Philippines, the burden of hypertension is both substantial and escalating, posing a severe challenge to the public health infrastructure. Dela Rosa et al. [12] highlight that despite various public health initiatives, the prevalence of uncontrolled hypertension remains alarmingly high. Their review of national surveillance data indicates persistent gaps in disease awareness, treatment adherence, and control rates among Filipinos. Specifically, lifestyle factors prevalent in the region, such as diets high in sodium and processed foods, coupled with increasing urbanization and sedentary behavior, exacerbate the risk profile of the population. This epidemiological reality underscores the urgent need for scalable, accessible screening tools that can identify high-risk individuals early in the disease trajectory, potentially before clinical symptoms manifest.

2.2 Machine Learning in Cardiovascular Risk Prediction

The limitations of traditional linear scoring systems, such as the Framingham Risk Score, have catalyzed a paradigm shift towards Machine Learning (ML) methodologies. Unlike traditional statistical methods, which often assume linear relationships between risk factors and outcomes, ML algorithms excel at modeling the high-dimensional, non-linear interactions inherent in complex physiological data.

2.2.1 The Shift from Linear to Non-Linear Models and Advanced Optimization

Recent literature demonstrates a clear trend towards the use of ensemble learning techniques for medical diagnosis, moving away from simple probabilistic models like Naive Bayes or standalone logistic regression models that may fail to capture complex biological dependencies. Naik et al. [3] and Pandey et al. [4] discuss how Artificial Intelligence (AI) and Digital Twins are reshaping hypertension prediction. Their findings suggest that algorithms capable of capturing complex feature interactions, specifically Random Forest and Gradient Boosting Machines (GBMs), consistently outperform these simpler baselines in terms of discrimination metrics like the Area Under the Receiver Operating Characteristic (AUC-ROC) curve.

However, maximizing the predictive capability of these complex ensembles requires rigorous hyperparameter tuning. Traditional exhaustive methods, such as Grid Search, often suffer from the "curse of dimensionality" when applied to algorithms like XGBoost or CatBoost, which possess vast parameter spaces. Recent methodological literature, specifically the foundational work of Bergstra and Bengio [13], demonstrates that Randomized Search is significantly more effective than Grid Search for high-dimensional optimization. By sampling the parameter space statistically, it can identify optimal hyperparameter regions with a fraction of the computational cost. Building upon this, modern implementations frequently adopt a Two-Stage approach: utilizing a broad randomized search for initial landscape exploration, followed by a concentrated local refinement around the best-performing candidates. This ensures a mathematically robust balance between exploration and exploitation.

2.2.2 Application on Survey Data and Class Imbalance

The utilization of national health surveys for training predictive models is a growing area of interest in health informatics. Elvas et al. [14] emphasize the utility of multimodal datasets that combine clinical history with anthropometric and lifestyle variables. In the Philippines, the National Nutrition Survey (NNS) provides a rich, albeit complex, data source. However, effectively utilizing such survey data requires robust preprocessing to handle challenges inherent to the methodology, such as missing values due to participant non-response, the high dimensionality of continuous dietary data, and severe class imbalance (where normotensive individuals significantly outnumber hypertensive ones).

Addressing this class imbalance is critical; standard algorithms biased towards the majority class produce superficially high accuracy but clinically dangerous false-negative rates. The machine learning literature advocates for two primary interventions to resolve this: algorithmic cost-sensitive learning and synthetic data generation. Applying Class Weights directly penalizes the misclassification of the minority class during optimization. Conversely, data-level interventions like the foundational Synthetic Minority Over-sampling Technique (SMOTE) [15], and its variant for mixed continuous and nominal data (SMOTENC), are frequently employed to expand the decision boundary of the minority class without the severe overfitting associated with naive random oversampling. Addressing these structural challenges through rigorous engineering interventions—such as optimal K-Nearest Neighbors imputation, strict collinearity fil-

tering, and robust resampling protocols—is central to ensuring model validity before calibration.

2.3 The Explainability Challenge: Explainable AI (XAI)

While the predictive power of “black-box” models such as Deep Neural Networks and XGBoost is undisputed, their opacity poses a significant barrier to clinical adoption. Mohapatra et al. [16] argue that in high-stakes medical environments, trust is contingent upon transparency. A clinician cannot ethically or practically act on a risk score without understanding the rationale behind the prediction, nor can they explain the risk to a patient to motivate lifestyle changes without intelligible insights. It is crucial to distinguish between interpretability, which refers to the extent to which a cause and effect can be observed natively within a system (often inherent in simple models like Decision Trees), and explainability, which involves using external techniques to elucidate the internal mechanics of complex black-box models. This study specifically seeks to extract explainability from high-performance ensemble models to bridge the gap between algorithmic accuracy and user trust.

To bridge this gap, Altukhi et al. [6] and Sreeja et al. [5] detail the rise of model-agnostic XAI methods. SHapley Additive exPlanations (SHAP), grounded in cooperative game theory, has become the gold standard for quantifying global and local feature importance. It provides a consistent way to attribute a prediction’s value to each feature, ensuring that the sum of feature contributions equals the model output. Similarly, Ribeiro et al. [17] introduced Local Interpretable Model-agnostic Explanations (LIME), which provides intuitive, perturbation-based explanations for individual instances by approximating the complex model with a simpler, interpretable one (like a linear model) locally around the prediction.

Beyond static explanations, the demand for actionable guidance has led to the development of counterfactual explanations. Wachter et al. [18] pioneered a framework emphasizing that explanations should not merely describe the current state, but prescribe the minimal perturbation required to alter an undesirable outcome. In medical domains, generating these “Wachter-style” counterfactuals is vital because it shifts XAI from descriptive analytics to prescriptive intervention, offering patients specific, realistic targets (e.g., reducing BMI by a precise, minimal amount) rather than abstract, unachievable metrics.

2.4 The Reliability Gap: Probability Calibration

A predictive model is considered calibrated if its predicted probabilities align with the true empirical frequency of events. For instance, if a model predicts a 30% risk of hypertension for a specific group of patients, exactly 30% of that group should truly be hypertensive in reality.

Löfström et al. [11, 8] identify a pervasive issue in modern ML: complex models like Deep Learning and Boosting ensembles tend to be poorly calibrated. Because these algorithms focus on minimizing classification error or maximizing the margin,

they often output overconfident probability scores (pushing probabilities towards 0 or 1). This results in a high Expected Calibration Error (ECE). In a clinical setting, an overconfident false alarm can lead to over-treatment and patient anxiety, while an underconfident miss can lead to negligence and a lack of early intervention. Therefore, calibration is not merely a statistical nicety but a safety requirement for decision support systems. Techniques such as Platt Scaling (fitting a sigmoid function) and Isotonic Regression (fitting a stepwise non-decreasing function) are standard post-hoc methods to correct this, while recent advancements like Venn-Abers predictors offer rigorous probability intervals that further quantify the empirical uncertainty of the prediction.

2.5 Synthesis and Research Gap

The intersection of these fields reveals a distinct research gap. While numerous studies, such as that by Eldem [9], apply ML to hypertension, and others like Niu et al. [19] apply XAI for transparency, there is a scarcity of systems that explicitly integrate probability calibration into the explanation pipeline, particularly for Philippine health data utilizing the National Nutrition Survey (NNS). Most existing systems explain the raw risk score, which poses a significant clinical danger: if the underlying model is uncalibrated, the XAI tool might convincingly explain a probability that is statistically incorrect. Furthermore, standard counterfactual generators often ignore the feasibility of the suggested changes, leading to mathematically optimal but clinically irrelevant recommendations.

This study addresses these gaps by adopting the Calibrated Explanations framework proposed by Löfström et al. [7], heavily augmented with Wachter-style counterfactual generation. By combining state-of-the-art classifiers—optimized via a rigorous Two-Stage Search—with post-hoc calibration methods and uncertainty-aware explanations, this research aims to produce a tool that is not only discriminatively accurate but also probabilistically reliable and transparent. Crucially, by integrating bounded optimization routines to ensure minimal clinical perturbation, this integration ensures that the explanations and counterfactuals provided to the user are mathematically grounded in corrected risk estimates and physiologically actionable, directly addressing the trust deficit in AI-driven healthcare.

3 Theoretical Framework

This section discusses the theoretical foundations and operational principles of the machine learning algorithms, sampling strategies, calibration methodologies, and explainability frameworks that will be used in this research. It provides the mathematical and conceptual basis for the proposed hypertension risk prediction system.

3.1 Heart Disease and Hypertension Modeling

Hypertension is the pathological state of sustained elevated blood pressure and is one of the most significant risk factors for cardiovascular disease. In the context of predictive modeling, the task will be formulated as a binary classification problem: mapping a vector of input features X encompassing clinical, anthropometric, lifestyle, and dietary variables to a target label $Y \in \{0, 1\}$, where 1 indicates the presence of hypertension and 0 indicates the absence of hypertension.

For this study, hypertension status will be defined using the adult diagnostic threshold stated in the 2020 Philippine Clinical Practice Guidelines, where a respondent will be classified as hypertensive if Systolic Blood Pressure (SBP) ≥ 140 mmHg or Diastolic Blood Pressure (DBP) ≥ 90 mmHg [1]. These values will be derived from the blood pressure measurements available in the Clinical and Health component of the DOST-FNRI National Nutrition Survey (NNS) [2].

The DOST-FNRI 2015 National Nutrition Survey will be used as the primary data source due to the need for culturally and locally relevant hypertension risk modeling. Existing hypertension models trained on foreign or homogeneous populations may not fully generalize to the Filipino population due to distinct dietary patterns, socioeconomic conditions, anthropometric profiles, and environmental factors. The NNS provides a nationally representative basis for modeling these local determinants, allowing the proposed system to learn from Philippine health and nutrition data.

3.2 Data Preprocessing

Raw survey data is rarely suitable for direct algorithmic ingestion. Rigorous preprocessing will therefore be required to standardize inputs, mitigate biases inherent in data collection, and prevent artificial performance inflation.

3.2.1 Target Definition and Leakage Guard

In predictive modeling, data leakage occurs when information from outside the intended training features is used by the model, or when the model is inadvertently given direct access to variables that define the target outcome. Since the binary Hypertension target will be mathematically derived from raw SBP and DBP readings, retaining these raw blood pressure variables as predictors would result in severe target leakage.

To prevent this, a leakage guard will be applied immediately after target creation. Direct and indirect blood-pressure-related variables will be removed from the feature set so that the model learns to estimate hypertension risk from clinical, anthropometric, behavioral, and dietary predictors rather than simply reconstructing the target label from the original blood pressure measurements.

3.2.2 Dimensionality Reduction via Collinearity Filtering

High dimensionality, particularly when features are highly correlated, can destabilize machine learning models and dilute the interpretability of feature attributions. This issue is especially relevant in nutritional and anthropometric datasets, where several variables may measure related physiological or dietary constructs.

To address this, a collinearity filter will be applied to the candidate feature space. Variables exhibiting a Pearson correlation coefficient greater than the selected threshold with another feature will be flagged for review. Redundant variables will be removed to reduce multicollinearity, improve computational efficiency, and support clearer explanations.

3.2.3 Handling Missing Values

Real-world survey data frequently contains missing entries due to non-response, measurement issues, or conditional questionnaire branching. To preserve data integrity and maximize usable sample size, this study will employ imputation strategies rather than simple row deletion whenever appropriate.

The quality of K-Nearest Neighbors (KNN) imputation is sensitive to the choice of k and to feature scaling. Since KNN relies on distance measurements, numerical variables with larger magnitudes can disproportionately influence neighbor selection. Therefore, numerical variables will be standardized before KNN-based imputation.

To determine the optimal number of neighbors, an artificial masking experiment will be conducted. A small percentage of observed non-missing values will be temporarily hidden, imputed under different candidate values of k , and compared against the original known values. The k value producing the lowest imputation error will be selected for the final preprocessing pipeline.

- **KNN Imputation:** For continuous numerical features, missing values will be estimated using K-Nearest Neighbors imputation. This method identifies similar respondents in the multidimensional feature space and imputes missing values based on nearby observations.
- **Mode Imputation:** For categorical variables, missing values will be imputed using the most frequent valid category, or the most frequent category among nearest neighbors when implemented within the KNN-based preprocessing workflow.

3.2.4 Feature Scaling

Feature scaling normalizes the range of independent variables. This step will be applied to continuous numerical features such as age, BMI, waist circumference, and dietary intake variables. Scaling is particularly important for distance-based algorithms and for KNN imputation, where unscaled variables with large magnitudes may dominate distance calculations.

- **Standardization (Z-score Scaling):** Standardization rescales a feature to have a mean μ of 0 and a standard deviation σ of 1:

$$X_{std} = \frac{X - \mu}{\sigma}$$

3.2.5 Class Imbalance Handling

Medical datasets frequently exhibit class imbalance, where the number of healthy or non-case individuals exceeds the number of individuals with the target condition. Standard algorithms may therefore bias predictions toward the majority class in order to maximize overall accuracy. This can be clinically dangerous because a model may appear accurate while missing many true hypertension-positive cases.

This study will evaluate several imbalance-handling configurations:

- **Baseline:** No resampling or weighting will be applied. This configuration will serve as the empirical control.
- **Class Weights (CW):** This technique modifies the algorithm's loss function to penalize misclassification of the minority class more heavily. It addresses imbalance directly within model optimization without altering the original data distribution.
- **SMOTE (Synthetic Minority Over-sampling Technique):** SMOTE synthesizes new minority instances by interpolating between existing minority samples in the feature space [15].
- **SMOTENC:** SMOTENC extends SMOTE for datasets containing both nominal and continuous variables.
- **Hybrid Approaches:** SMOTE + Class Weights and SMOTENC + Class Weights will be evaluated to determine whether synthetic generation combined with cost-sensitive learning improves model performance.

3.3 Machine Learning Algorithms

This study will evaluate a diverse suite of supervised learning algorithms, ranging from interpretable baselines to complex ensemble models. A comparative approach using eight algorithms will be used to establish a rigorous performance baseline and identify the most suitable model family for the NNS hypertension prediction task.

3.3.1 Linear and Instance-Based Models

Logistic Regression (LR) Logistic Regression is a foundational statistical model for binary classification. It estimates the probability that an instance belongs to a particular class using the logistic sigmoid function. For a feature vector $X = (x_1, \dots, x_n)$, the predicted probability is calculated as:

$$P(y = 1|X) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \dots + \beta_n x_n)}}$$

Although Logistic Regression is mathematically linear in its parameters, it will serve as an essential interpretable baseline against which more complex non-linear models can be compared.

k-Nearest Neighbors (kNN) k-Nearest Neighbors is a non-parametric, instance-based learner. It classifies a new instance based on the majority vote of its k closest neighbors in the feature space. Proximity is commonly computed using Euclidean distance:

$$d(p, q) = \sqrt{\sum_{i=1}^n (q_i - p_i)^2}$$

kNN will be included because it captures local similarity among patient profiles and provides a useful comparison to more complex global decision models.

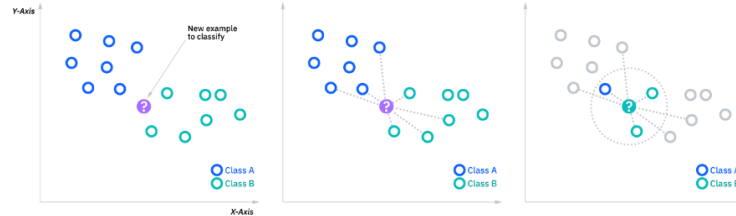


Figure 1: Example of k-Nearest Neighbors (kNN) classification. (Source: [20])

Naive Bayes (NB) Naive Bayes is a probabilistic classifier based on Bayes' Theorem. It assumes conditional independence between features given the class label. Although this assumption may not fully hold in complex medical data, Naive Bayes is computationally efficient and provides a useful probabilistic baseline for evaluating more advanced models and calibration behavior.

3.3.2 Ensemble Methods

Ensemble methods combine multiple base learners to improve generalizability and robustness over a single estimator.

Random Forest (RF) Random Forest is a bagging ensemble that constructs multiple decision trees during training and outputs the class selected by the majority of the trees. It is resistant to overfitting through bootstrap aggregation and feature randomness. It will be used as a strong tree-based benchmark for structured survey data.

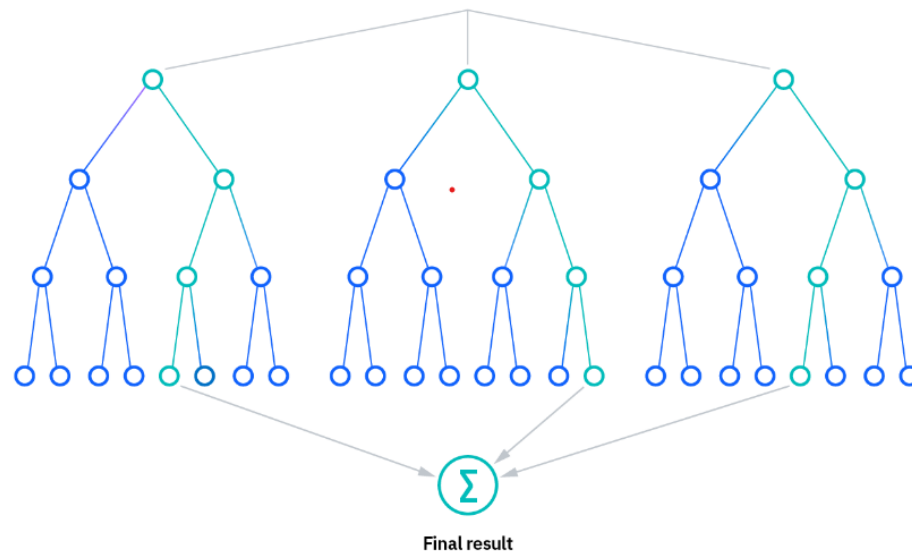


Figure 2: Diagram of a Random Forest (RF) model. (Source: [21])

Adaptive Boosting (AdaBoost) AdaBoost is a boosting technique that trains learners sequentially. Each subsequent learner focuses more strongly on instances misclassified by previous learners by increasing their weights. This iterative error-correction mechanism can improve performance on hard-to-classify cases.

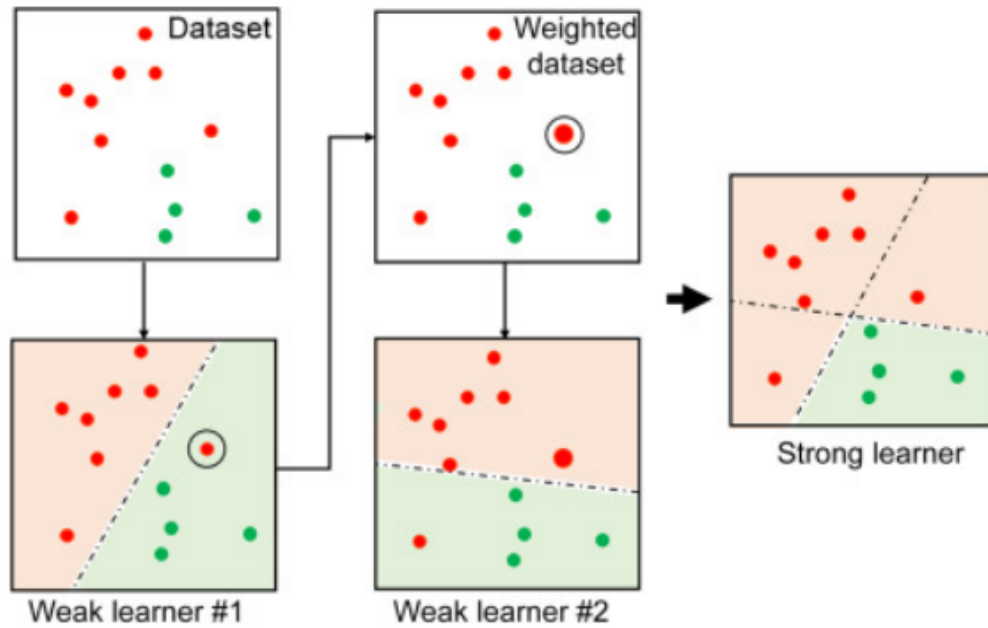


Figure 3: Conceptual model of Adaptive Boosting (AdaBoost). (Source: [22])

XGBoost (Extreme Gradient Boosting) XGBoost is an optimized gradient boosting framework that builds trees sequentially to minimize a regularized objective function. It is widely used for structured tabular datasets due to its performance, regularization capabilities, and ability to capture non-linear feature interactions. It will be included as one of the major candidate algorithms for hypertension prediction.

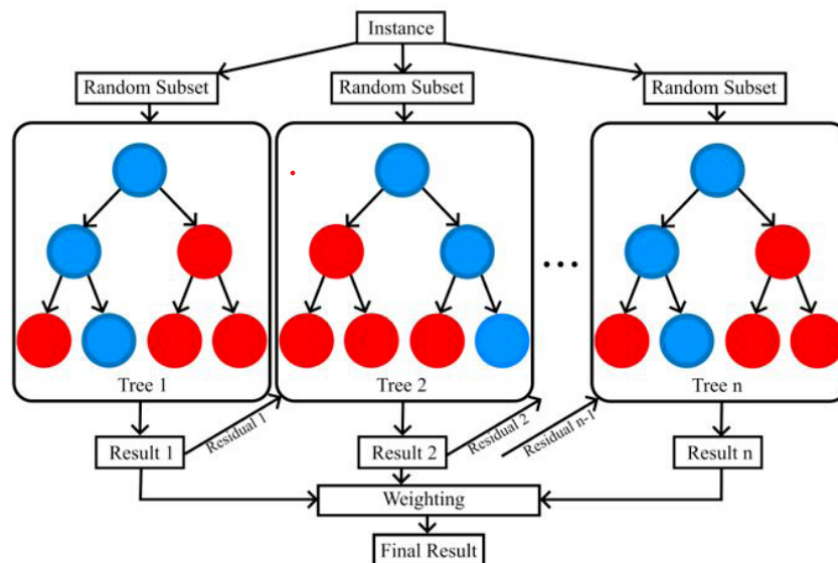


Figure 4: Simplified structure of an XGBoost model. (Source: [23])

LightGBM and CatBoost LightGBM and CatBoost are advanced gradient boosting variants. LightGBM uses leaf-wise tree growth and histogram-based learning, making it efficient for large datasets. CatBoost implements ordered boosting and provides support for categorical variables. Both will be evaluated as high-performing ensemble candidates for structured health survey data [24, 25].

3.4 Two-Stage Rigorous Hyperparameter Optimization

To maximize model performance while managing computational cost, this study will use a Two-Stage Rigorous Optimization workflow informed by the efficiency of randomized parameter search [13]. Instead of relying on exhaustive grid search, the study will sample from defined statistical distributions for important hyperparameters.

- **Stage 1 (Broad Exploration):** A broad Randomized Search will be conducted across the allowable hyperparameter space of each model. This stage will identify promising regions of the search space while avoiding the computational burden of exhaustive grid search.
- **Stage 2 (Local Refinement):** The strongest configurations identified in Stage 1 will be subjected to a narrower local refinement search. This phase will concentrate around promising hyperparameter regions to fine-tune parameters such as learning rate, depth, regularization, and sampling ratios.
- **Final Training and Validation Evaluation:** The best candidate configurations will be trained on the training split and evaluated on validation and test partitions. Early stopping or equivalent validation-based safeguards will be used when applicable to reduce overfitting.

3.5 Model Evaluation Framework

To assess both the discriminative ability and probability reliability of the candidate models, this study will employ a combination of classification and calibration metrics.

3.5.1 Discrimination Metrics

These metrics evaluate the ability of the model to distinguish between hypertensive and normotensive individuals.

- **Accuracy:** The proportion of correct predictions among all predictions:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

where TP is true positives, TN is true negatives, FP is false positives, and FN is false negatives.

- **Recall (Sensitivity):** The proportion of actual hypertensive cases correctly identified:

$$Recall = \frac{TP}{TP + FN}$$

In medical screening, high recall is important because it minimizes missed positive cases.

- **Precision:** The proportion of predicted hypertensive cases that are actually positive:

$$Precision = \frac{TP}{TP + FP}$$

- **F1-Score:** The harmonic mean of precision and recall:

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

- **Area Under the ROC Curve (AUC):** AUC measures the ability of the model to rank positive cases above negative cases across classification thresholds. Values closer to 1.0 indicate stronger discrimination.

3.5.2 Calibration Metrics

These metrics evaluate whether predicted probabilities are reliable.

- **Log Loss:** Log Loss penalizes confident but incorrect predictions:

$$LogLoss = -\frac{1}{N} \sum_{i=1}^N [y_i \log(p_i) + (1 - y_i) \log(1 - p_i)]$$

where p_i is the predicted probability for the positive class.

- **Expected Calibration Error (ECE):** ECE measures the difference between predicted confidence and empirical accuracy across probability bins. Let B_m denote the set of samples in bin m :

$$acc(B_m) = \frac{1}{|B_m|} \sum_{i \in B_m} \mathbf{1}(y_i = \hat{y}_i)$$

$$conf(B_m) = \frac{1}{|B_m|} \sum_{i \in B_m} \hat{p}_i$$

$$ECE = \sum_{m=1}^M \frac{|B_m|}{N} |acc(B_m) - conf(B_m)|$$

Lower ECE indicates better calibration.

3.6 Probability Calibration

Calibration addresses the reliability of predicted probabilities. A model is considered calibrated if the predicted probabilities align with observed empirical frequencies. This study will evaluate post-hoc probability calibration methods to address the reliability gap commonly observed in complex machine learning models.

- **Platt Scaling:** Platt Scaling fits a logistic regression model to raw classifier scores. It produces a calibrated probability $\hat{P}$ using:

$$\hat{P}(y = 1|f(x)) = \frac{1}{1 + \exp(Af(x) + B)}$$

where A and B are learned calibration parameters.

- **Isotonic Regression:** Isotonic Regression is a non-parametric calibration method that fits a monotonic function g to transform raw model scores into calibrated probabilities:

$$\min_g \sum_{i=1}^N (y_i - g(f_i))^2$$

subject to the constraint that g is monotonically non-decreasing.

- **Venn-Abers Predictors:** Venn-Abers predictors are based on conformal prediction principles and produce probability intervals. For a test instance with score f_{test} , two isotonic regressions are fitted under the assumptions of class 0 and class 1:

$$p_0 = g_0(f_{test})$$

$$p_1 = g_1(f_{test})$$

A single probability point estimate can be derived using:

$$p = \frac{p_1}{1 - p_0 + p_1}$$

This produces a probability interval that provides a more informative representation of predictive uncertainty than a single uncalibrated point estimate.

3.7 Explainable Artificial Intelligence (XAI)

Explainable Artificial Intelligence methods will be used to make the model outputs more interpretable to users. In this study, XAI will be used to provide both global feature importance and patient-specific explanations.

3.7.1 SHAP (SHapley Additive exPlanations)

SHAP assigns each feature an importance value for a particular prediction [26]. Based on Shapley values from cooperative game theory, the contribution of feature i is computed as:

$$\phi_i = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!} [f_x(S \cup \{i\}) - f_x(S)]$$

where F is the set of all features and S is a subset excluding feature i . SHAP will be used for both global attribution and local explanation.

3.7.2 LIME (Local Interpretable Model-agnostic Explanations)

LIME explains individual predictions by learning a simpler interpretable model locally around the prediction. It solves:

$$\text{explanation}(x) = \arg \min_{g \in G} \mathcal{L}(f, g, \pi_x) + \Omega(g)$$

where $\mathcal{L}$ measures local unfaithfulness, π_x defines locality around instance x , and $\Omega(g)$ penalizes explanation complexity [17]. LIME will be used to provide accessible local explanations for users.

3.7.3 The Calibrated Explanations Framework

This study will integrate probability calibration with explanation generation using the Calibrated Explanations framework proposed by Löfström et al. [7]. Instead of explaining only raw model outputs, the system will generate explanations grounded in calibrated risk estimates. This will help reduce the risk of presenting convincing explanations for unreliable or overconfident probability scores.

3.7.4 Counterfactual Explanations via Wachter-Style Bounded Optimization

The study will generate counterfactual explanations following the prescriptive principles introduced by Wachter et al. [18]. The goal is to estimate the smallest realistic feature change that would shift a high-risk prediction below the selected screening boundary.

For each actionable continuous feature, bounded Brent's method through `scipy.optimize.minimize_scalar` will be used to minimize:

$$L(x') = \lambda \cdot (f(x') - y_{target})^2 + \frac{(x' - x)^2}{\sigma^2}$$

where:

- $f(x')$ is the model's predicted probability at the proposed counterfactual value x' .
- y_{target} will be set slightly below the decision threshold selected during validation.

- λ controls the trade-off between crossing the decision boundary and keeping the change minimal.
- $\frac{(x'-x)^2}{\sigma^2}$ is a variance-normalized proximity penalty that accounts for the scale of each feature.

Unlike naive grid-sweep methods, bounded Brent's optimization will efficiently search for the optimal single-feature change within the physiological bounds of that variable.

4 Design and Implementation

This section presents the proposed realization of the theoretical framework into a functioning software system. It describes the system architecture, dataset composition, data engineering pipeline, model development plan, hardware requirements, and proposed web application interface.

4.1 System Framework

The proposed system will implement a full-stack data science pipeline. The workflow will be structured into sequential phases: Data Ingestion, Preprocessing, Model Training, Probability Calibration, Explainability, Counterfactual Generation, and Web Application Deployment. This pipeline will ensure that raw survey data is systematically transformed into interpretable and probability-aware hypertension risk estimates.

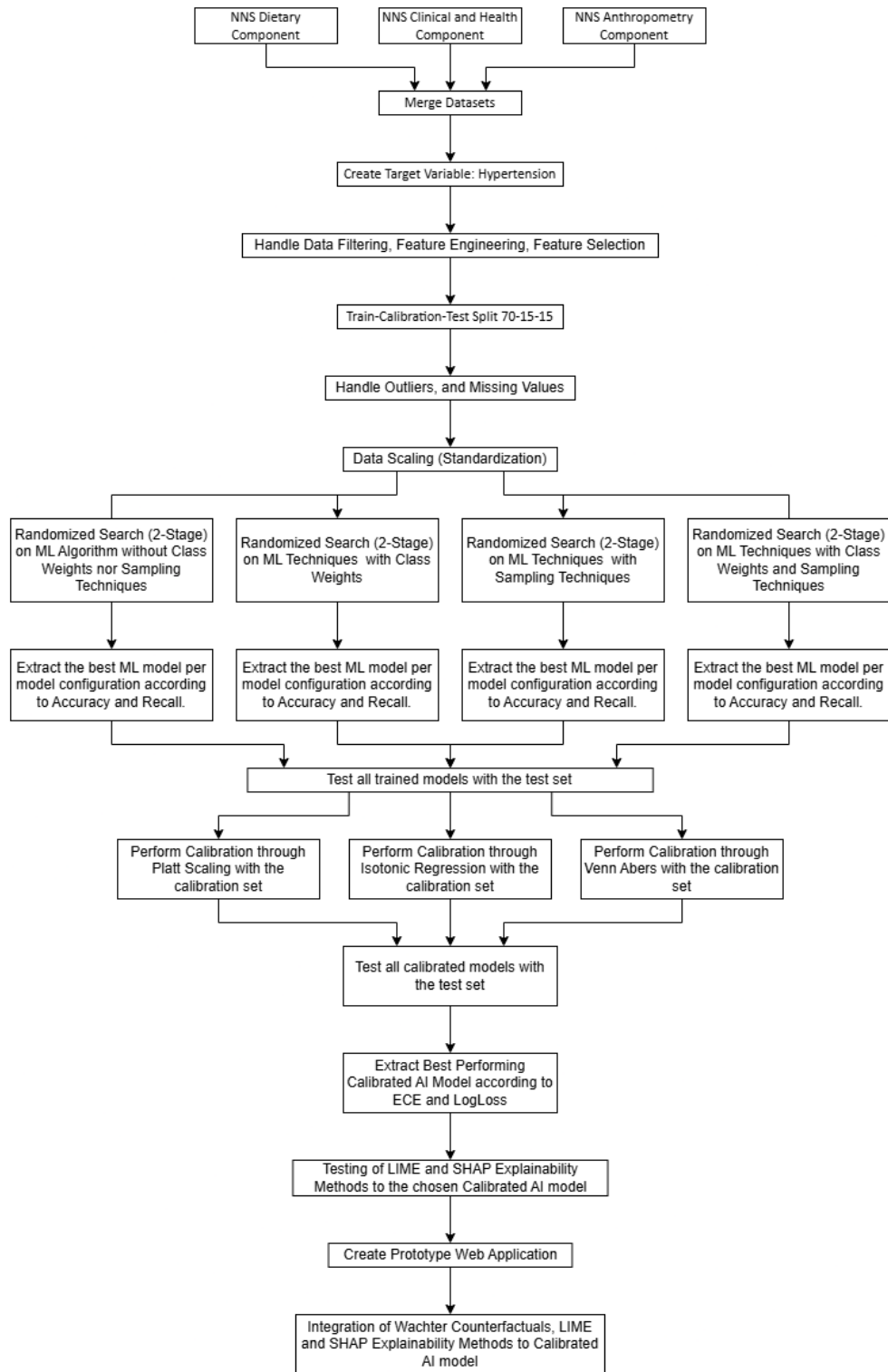


Figure 5: Proposed Methodological Workflow Diagram.

4.2 Data Source and Variable Definitions

4.2.1 The DOST-FNRI 2015 NNS Dataset

The study will utilize microdata from the 2015 National Nutrition Survey (NNS) conducted by the Department of Science and Technology - Food and Nutrition Research Institute (DOST-FNRI). To create a comprehensive feature set for hypertension modeling, three survey components will be merged using unique respondent identifiers:

1. **Clinical and Health Component:** This will provide blood pressure measurements for target-variable construction and behavioral risk variables such as smoking and alcohol consumption.
2. **Anthropometric Component:** This will provide physical body measurements required for calculating indices such as BMI, waist circumference, and hip circumference.
3. **Individual Dietary Component:** This will provide food intake and nutrient information that will be aggregated into dietary feature groups and nutrient totals.

4.2.2 Candidate Data Variables

The analytical dataset is expected to include clinical, anthropometric, lifestyle, dietary, and engineered variables, including one binary hypertension target variable. Tables 1, 2, and 3 list the candidate variables that will be considered for modeling. Some variables may be removed during preprocessing, leakage control, collinearity filtering, or feature selection.

Table 1: Candidate Variables for Hypertension Prediction Model: Clinical, Anthropometric, and Lifestyle Features

Variable Name	Role	Description
Hypertension	Target	Binary outcome variable to be constructed using the adult threshold of SBP $\geq$ 140 mmHg or DBP $\geq$ 90 mmHg.
age	Feature	Age of respondent in years.
sex	Feature	Biological sex.
ethnicity	Feature	Ethnicity code available in the survey.
pa_met	Feature	Physical activity expressed as metabolic equivalent.
fbs	Feature	Fasting blood sugar.
chol	Feature	Total cholesterol.
tri	Feature	Triglycerides.
hdl	Feature	High-density lipoprotein.
ldl	Feature	Low-density lipoprotein.
waist	Feature	Waist circumference.
hip	Feature	Hip circumference.
BMI	Feature	Body Mass Index to be calculated from height and weight.
fe_alcohol_level	Feature	Engineered alcohol-use level derived from survey alcohol variables.
fe_smoking_level	Feature	Engineered smoking level derived from survey smoking variables.

Table 2: Candidate Variables for Hypertension Prediction Model: Dietary Food Groups

Variable Name	Role	Description
epwt_fg1	Feature	Cereals and cereal products.
epwt_fg2	Feature	Rice and rice products.
epwt_fg3	Feature	Corn and corn products.
epwt_fg4	Feature	Other cereal products.
epwt_fg5	Feature	Starchy roots and tubers.
epwt_fg6	Feature	Sugar and syrups.
epwt_fg7	Feature	Dried beans, nuts, and seeds.
epwt_fg8	Feature	Vegetables.
epwt_fg9	Feature	Green leafy and yellow vegetables.
epwt_fg10	Feature	Other vegetables.
epwt_fg11	Feature	Fruits.
epwt_fg12	Feature	Vitamin C-rich fruits.
epwt_fg13	Feature	Other fruits.
epwt_fg14	Feature	Fish, meat, and poultry.
epwt_fg15	Feature	Fish and fish products.
epwt_fg16	Feature	Meat and meat products.
epwt_fg17	Feature	Poultry.
epwt_fg18	Feature	Eggs.
epwt_fg19	Feature	Milk and milk products.
epwt_fg20	Feature	Whole milk.
epwt_fg21	Feature	Other milk products.
epwt_fg23	Feature	Fats and oils.
epwt_fg24	Feature	Miscellaneous.
epwt_fg25	Feature	Beverages.
epwt_fg26	Feature	Condiments and spices.
epwt_fg27	Feature	Other miscellaneous items.

Table 3: Candidate Variables for Hypertension Prediction Model: Nutrient Totals

Variable Name	Role	Description
Total_Food_epwt	Feature	Total food weight.
Total_Ener	Feature	Total energy intake.
Total_Prot	Feature	Total protein intake.
Total_Calc	Feature	Total calcium intake.
Total_Iron	Feature	Total iron intake.
Total_VitA	Feature	Total vitamin A intake.
Total_VitC	Feature	Total vitamin C intake.
Total_Thia	Feature	Total thiamin intake.
Total_Ribo	Feature	Total riboflavin intake.
Total_Nia	Feature	Total niacin intake.
Total_CHO	Feature	Total carbohydrate intake.
Total_Fat	Feature	Total fat intake.

4.3 Data Preprocessing and Feature Engineering

This section describes the planned data transformation pipeline that will be applied to the raw NNS microdata to ensure data quality, relevance, and model readiness.

4.3.1 Data Merging Strategy

The study will integrate the Clinical and Health, Anthropometric, and Individual Dietary datasets to create a unified respondent profile. The merge will be performed using respondent-level identifiers such as household number and member code. The Clinical and Health component will serve as the base table to ensure that retained respondents have available blood pressure measurements for target construction.

4.3.2 Data Cleaning and Filtering

To align the dataset with the study's scope and clinical guidelines, the following filtering steps will be applied:

- **Dropping Invalid Target Entries:** Records without sufficient systolic or diastolic blood pressure data will be removed because hypertension status cannot be established for those individuals.
- **Exclusion of Non-Relevant Anthropometric Groups:** Respondents outside the intended adult modeling scope, such as infants, children, and special physiological groups, will be excluded when their diagnostic interpretation requires different medical guidelines.
- **Pregnant and Lactating Women:** Pregnant and lactating respondents will be excluded due to confounding physiological factors that require specialized medical interpretation distinct from the general adult population.
- **Adult Modeling Scope:** The final analytical dataset will focus on adult respondents for whom the fixed adult hypertension threshold of $\geq 140/90$ mmHg is clinically applicable.

4.3.3 Feature Engineering

New variables will be constructed to capture clinically relevant risk factors:

- **Target Variable (Hypertension):** A binary target variable will be engineered based on the adult hypertension threshold. A respondent will be labeled hypertensive if average SBP ≥ 140 mmHg or average DBP ≥ 90 mmHg; otherwise, the respondent will be labeled non-hypertensive.
- **Body Mass Index (BMI):** BMI will be calculated from weight and height measurements using the standard formula.
- **Lifestyle Levels and Cascading Fallbacks:** Smoking and alcohol variables will be transformed into cleaner categorical levels. If direct aggregate variables are missing or coded as not applicable, supporting raw survey variables will be used to infer the most appropriate category.

- **Dietary Aggregation:** Food intake variables will be organized into food groups and nutrient totals to produce a more interpretable dietary feature set.

4.3.4 Handling Missing Values and Outliers

- **Outlier Detection:** Numerical features will be screened for biologically implausible or structurally invalid values. Such values will be treated as missing or excluded depending on the severity and context.
- **Standardization for Imputation:** Numerical variables will be standardized before KNN imputation to prevent large-scale variables from dominating Euclidean distance calculations.
- **Optimization of Imputation Parameters:** A masking experiment will be conducted to determine the optimal number of neighbors k for KNN imputation. Candidate values of k will be evaluated by comparing imputed masked values against their known original values.
- **KNN Imputation Application:** Using the selected value of k , missing numerical values will be imputed using neighboring observations, while categorical values will be imputed using the most frequent valid category.

4.3.5 Feature Selection

To reduce dimensionality and noise, redundant variables will be removed:

- **Identifiers:** Administrative identifiers and variables not intended as clinical predictors will be removed.
- **Redundant Dietary Weights:** Raw dietary weight variables may be removed in favor of edible-portion equivalents or aggregated dietary variables.
- **Origin Variables:** Intermediate variables used only to engineer lifestyle levels will be removed after feature construction to reduce redundancy and prevent leakage.
- **Collinearity Filtering:** Highly correlated variables will be evaluated and reduced to preserve the most clinically meaningful and computationally useful predictors.

4.4 Model Development and Optimization Strategy

The study will employ a rigorous, multi-stage model development workflow to identify the most suitable hypertension prediction model.

4.4.1 Stage 1: Randomized Search and Hyperparameter Spaces

A Randomized Search with Stratified K-fold Cross-Validation will be performed across eight algorithms: XGBoost, CatBoost, LightGBM, Random Forest, AdaBoost, Logistic Regression, k-Nearest Neighbors, and Naive Bayes. Table 4 summarizes the planned hyperparameter search spaces.

Table 4: Planned Hyperparameter Search Space Distributions for Two-Stage Optimization

Algorithm	Hyperparameter	Search Space Boundary / Distribution
Logistic Regression	C	Log-Uniform [0.0001, 100.0]
	solver	Categorical [lbfgs, liblinear]
k-Nearest Neighbors	n_neighbors	Integer [3, 30]
	weights	Categorical [uniform, distance]
	metric	Categorical [euclidean, manhattan]
Naive Bayes	var_smoothing	Log-Uniform [1e-12, 1e-8]
AdaBoost	n_estimators	Integer [50, 300]
	learning_rate	Uniform [0.01, 0.50]
	base_depth	Integer [1, 4]
Random Forest	max_features	Uniform [0.2, 1.0]
	max_depth	Categorical [None, 10, 20, 30]
	min_samples_split	Integer [2, 19]
	min_samples_leaf	Integer [1, 9]
XGBoost	learning_rate	Log-Uniform [0.01, 0.30]
	max_depth	Integer [3, 8]
	subsample / colsample_bytree	Uniform [0.6, 1.0]
	reg_lambda	Log-Uniform [0.01, 10.0]
LightGBM	learning_rate	Log-Uniform [0.01, 0.30]
	max_depth	Integer [3, 9]
	num_leaves	Integer [15, 149]
	subsample / colsample_bytree	Uniform [0.6, 1.0]
CatBoost	learning_rate	Log-Uniform [0.01, 0.30]
	depth	Integer [4, 9]
	l2_leaf_reg	Uniform [1.0, 10.0]
	random_strength	Uniform [0.1, 2.0]

- **Configurations to be Tested:** Each algorithm will be evaluated across the planned imbalance-handling configurations: Baseline, Class Weights, SMOTE, SMOTENC, SMOTE + Class Weights, and SMOTENC + Class Weights.
- **Selection Criteria:** During Stage 1, top models will be identified using discrimination metrics such as F1-Score and AUC. These metrics will be used to identify models that balance classification utility and ranking ability.

4.4.2 Stage 2: Secondary Selection for Calibration

From the pool of top configurations identified in Stage 1, a second filtering stage will be applied to determine which models will proceed to calibration. This stage will prioritize models with strong Recall, competitive Accuracy, and stable AUC, since the system is intended for screening where missed positive cases are clinically important.

4.4.3 Stage 3: Probability Calibration and Final Selection

The selected models will undergo post-hoc calibration using Platt Scaling, Isotonic Regression, and Venn-Abers predictors.

- **Final Selection Criteria:** The calibrated models will be evaluated using a clinically oriented hierarchy rather than a single metric alone. Recall will be prioritized to preserve hypertension-positive case detection, AUC will be used to verify discriminative ability, and ECE and Log Loss will be used to assess probability reliability after calibration. The final model will be selected based on its overall suitability for screening deployment, balancing sensitivity, discrimination, calibration quality, and compatibility with the explainability pipeline.

4.4.4 Stage 4: Explainability and Counterfactual Generation

After the final calibrated model is selected, explanation and counterfactual modules will be integrated.

- **Feature Masking for Actionability:** Immutable variables such as age, sex, ethnicity, and height will be marked as non-actionable. Variables such as BMI, dietary intake, and smoking or alcohol levels will be considered actionable or partially actionable depending on clinical plausibility.
- **Counterfactual Search via Bounded Optimization:** Bounded Brent's method will be used to search for the minimal feasible perturbation of actionable features required to reduce predicted risk below the selected screening threshold. A variance-normalized penalty will be applied to favor smaller, realistic lifestyle changes.

4.4.5 Modular Notebook-Based Training Workflow

For computational tractability, the model development workflow will be modularized into separate experiment notebooks grouped by algorithm family. This will allow long-running model searches to be executed independently while preserving a consistent preprocessing, optimization, and evaluation protocol across all candidates. A dedicated compilation notebook will aggregate the exported results from the algorithm-specific notebooks to produce the final comparison tables for analysis.

4.5 System Design

4.5.1 Context Diagram

The proposed system will operate as a web-based application where the user provides health parameters and receives a risk assessment output. The core process will

involve input validation, preprocessing, model inference, probability calibration, explanation generation, and counterfactual generation. The system will access a stored model pipeline containing the selected trained model, preprocessing components, and calibration module.

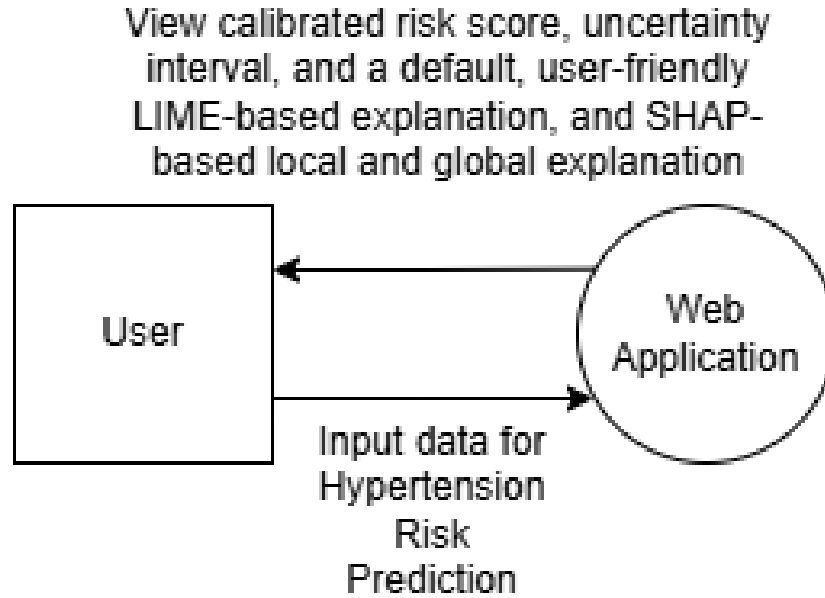


Figure 6: Proposed System Context Diagram.

4.5.2 Use-Case Diagram

Figure 7 illustrates the proposed functional requirements of the system. The primary actor, representing either a patient, clinician, or health-conscious user, will interact with the system to perform the following actions:

- **Input Health Data:** The user will enter demographic, anthropometric, clinical, lifestyle, and dietary information.
- **View Risk Assessment:** The user will view a calibrated hypertension risk estimate and, when applicable, an uncertainty interval.
- **View Explanations:** The user will view local and global explanations generated through LIME and SHAP.
- **View Actionable Counterfactuals:** The user will review lifestyle-oriented counterfactual suggestions generated by the optimization module.

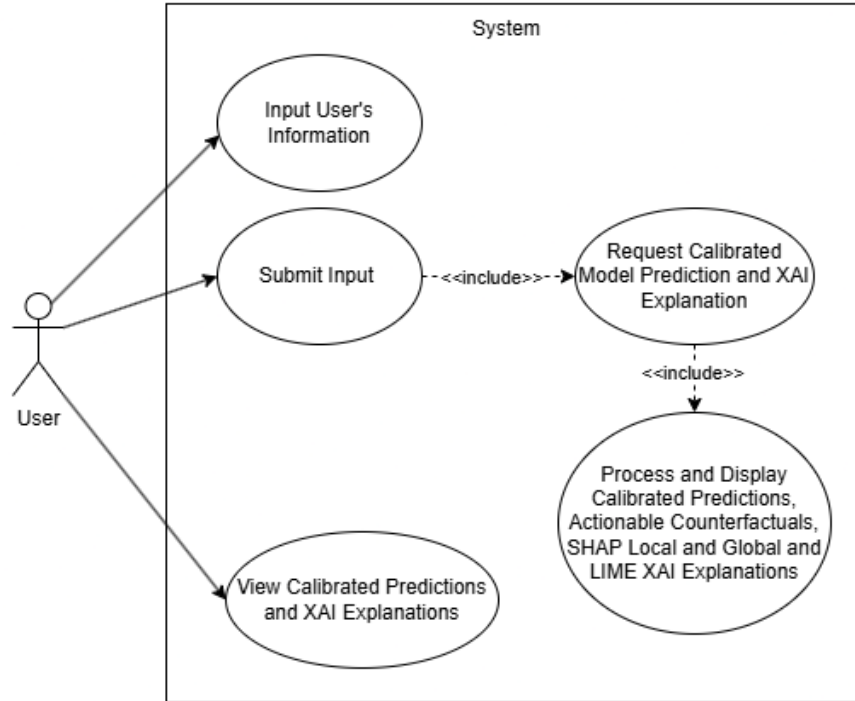


Figure 7: Proposed System Use Case Diagram.

4.5.3 Activity Diagram

The proposed operational flow is shown in Figure 8. The system workflow will proceed as follows:

1. The user will input health and lifestyle data through the web interface.
2. The system will validate and preprocess the input.
3. The selected machine learning model will generate a raw prediction score.
4. The calibration layer will transform the raw score into a calibrated probability estimate.
5. The system will generate local explanations and counterfactual suggestions.
6. The frontend will display the risk estimate, uncertainty information, explanations, and recommended counterfactual changes.
7. If the user requests advanced interpretation, the system will display additional SHAP-based visualizations.

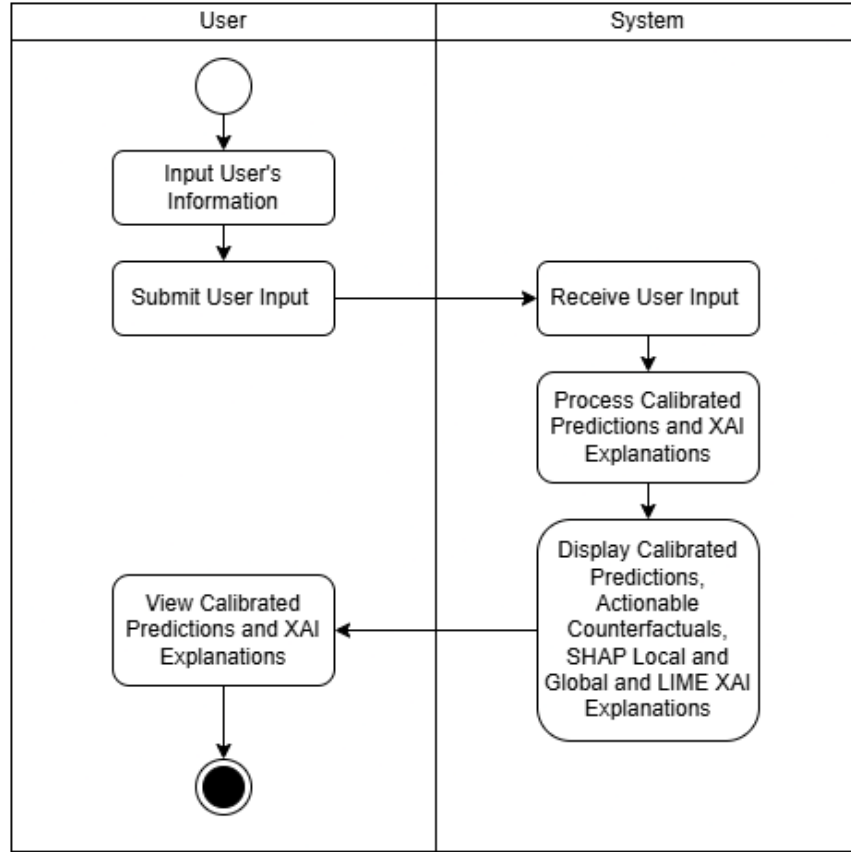


Figure 8: Proposed System Activity Diagram.

4.6 Technical Architecture

The proposed system will be built on a modular technical stack designed to support both computationally intensive model development and lightweight web application usage.

4.6.1 System Architecture

The software architecture will follow a modular design:

- **Backend:** The backend will be developed in Python, using machine learning and data processing libraries such as scikit-learn, XGBoost, LightGBM, CatBoost, SHAP, LIME, and calibration-related packages.
- **Frontend:** The frontend will be built using Streamlit to support rapid development of interactive visualizations and input forms.
- **Model Persistence:** Trained preprocessing pipelines, selected models, and calibration components will be serialized using `joblib` or equivalent persistence tools.

4.6.2 Hardware Requirements for Model Training

The training phase will involve computationally intensive tasks such as randomized hyperparameter search, ensemble model training, calibration, and explanation generation. The recommended development environment will include:

- **Processor:** Multi-core CPU with at least 8 cores for parallel processing.
- **Memory (RAM):** At least 16 GB RAM to support in-memory processing of the merged NNS dataset.
- **GPU:** A CUDA-compatible GPU is recommended to accelerate supported model training and parallel computation.
- **Storage:** SSD storage with sufficient free space for raw datasets, processed outputs, model artifacts, and generated result files.

4.6.3 Requirements for Web Application Usage

The deployed prototype will be designed to be lightweight and accessible through standard consumer devices.

- **Device:** Any modern computer, tablet, or smartphone.
- **Operating System:** Platform-independent access through Windows, macOS, Linux, Android, or iOS.
- **Web Browser:** A modern web browser with JavaScript enabled.
- **Internet Connection:** A stable internet connection will be required if the application is hosted remotely.

5 Expected Output

This section presents the expected outputs of the proposed HRP-AI system. The web application will be designed as a prototype decision-support tool that allows users to enter clinical, anthropometric, lifestyle, and dietary information, receive a calibrated hypertension risk estimate, view model explanations, and inspect actionable counterfactual suggestions. The following sample screens represent the major processes expected in the final system.

5.1 Landing Page

The landing page will serve as the entry point of the web application. It will introduce the purpose of the system, provide a brief context on hypertension screening, and guide the user toward the risk assessment module through a clear call-to-action button.

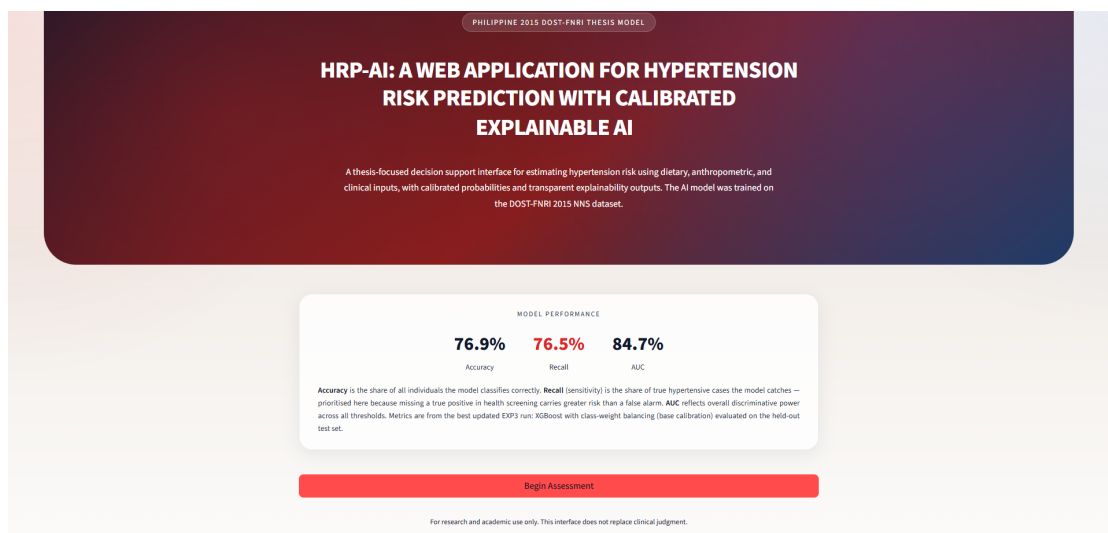


Figure 9: Expected Landing Page of the HRP-AI Web Application

5.2 Progressive Input Interface

The user input module will be structured as a progressive multi-step form to reduce cognitive load and organize data entry into logical sections. The form will collect clinical profile information, anthropometric measurements, lifestyle variables, and dietary intake values.

Hypertension Risk Assessment

Back Home

Patient Details

Complete all sections below. Click Predict My Risk to see the output below this form on the same page.

Enter the requested values below. Hover the ? icon beside each field to view hints and definitions.

Quick Identity and Key Inputs

Age (years)

missing

Sex

missing

Ethnicity

0 - No, Not an IP/Without Foreign Blood (default)

Behavioral (Smoking and Alcohol)

Provide original smoking and alcohol variables. The backend computes engineered levels.

Smoking Status

missing

Current Smoking Frequency

missing

Smoking History

missing

Alcohol Use Status

missing

Alcohol Consumption

missing

Alcohol Use in Past 12 Months

missing

Alcohol Use in Past 30 Days

missing

Binge Drinking Status

missing

Anthropometrics

For engineered BMI and WHR, provide these origin anthropometric measurements.

Weight (kg)

Height (cm)

Waist Circumference (cm)

Figure 10: Expected Input Step 1: Foundational Clinical and Biological Profile

Weight (kg)

0.0

Height (cm)

0.0

Waist Circumference (cm)

0.0

Hip Circumference (cm)

0.0

Dietary

Enter each dietary value as your daily intake. Hover the ? icon on each field for definitions and examples.

Rice and Rice Products (in grams)

0

Corn and Corn Products (in grams)

0

Other Cereal Products (in grams)

0

Cereals and Cereal Products (in grams)

0.00

Starchy Roots and Tubers (in grams)

0

Sugar and Syrups (in grams)

0

Auto-summed from the food-group inputs above.

Dried Beans

0

Green Leafy and Yellow (in grams)

0

Other Vegetables (in grams)

0

Vegetables (in grams)

0.00

Vitamin C-Rich Fruits (in grams)

0

Other Fruits (in grams)

0

Auto-summed from the food-group inputs above.

Fruits (in grams)

0.00

Fish and Fish Products (in grams)

0

Meat and Meat Products (in grams)

0

Auto-summed from the food-group inputs above.

Poultry (in grams)

Eggs (in grams)

Fish, Meat and Poultry (in grams)

Figure 11: Expected Input Step 2: Anthropometric Measurements with Dynamic BMI Calculation

The anthropometric section will include automatic computation of Body Mass Index (BMI) based on height and weight inputs. This will reduce manual calculation errors and ensure that the model receives internally consistent input values.

Poultry (in grams) 0	Eggs (in grams) 0	Fish, Meat and Poultry (in grams) 0.00 Auto-summed from the food-group inputs above.
Whole Milk (in grams) 0	Milk Products (in grams) 0	Milk and Milk Products (in grams) 0.00 Auto-summed from the food-group inputs above.
Fats and Oils (in grams) 0	Beverages (in grams) 0	Condiments and Spices (in grams) 0
Other Miscellaneous (in grams) 0	Miscellaneous (in grams) 0.00 Auto-summed from the food-group inputs above.	Total Calcium (mg) 0
Total Carbohydrates (g) (g/day) 0	Total Fats (g) (g/day) 0	Total Iron (mg) 0.0
Total Niacin (mg) 0	Total Riboflavin (mg) 0	Total Thiamin (mg) 0
Total Vitamin A (mcg RE) (mcg RE/day) 0	Total Vitamin C (mg) 0	
Total Food Intake (g) 0.00 Auto-updates from the dietary values above.	Total Energy (kcal) 0.00 Auto-updates from the dietary values above.	Total Protein (g) 0.00 Auto-updates from the dietary values above.

Figure 12: Expected Input Step 3: Behavioral Risk Factors and Lifestyle Profiling

Total Carbohydrates (g) (g/day) 0	Total Fats (g) (g/day) 0	Total Iron (mg) 0.0
Total Niacin (mg) 0	Total Riboflavin (mg) 0	Total Thiamin (mg) 0
Total Vitamin A (mcg RE) (mcg RE/day) 0	Total Vitamin C (mg) 0	
Total Food Intake (g) 0.00 Auto-updates from the dietary values above.	Total Energy (kcal) 0.00 Auto-updates from the dietary values above.	Total Protein (g) 0.00 Auto-updates from the dietary values above.

Complete all missing inputs before predicting: Age, Sex, Smoking Status, Current Smoking Frequency, Smoking History, Alcohol Use Status, Alcohol Consumption, Alcohol Use in Past 12 Months, and 2 more.

Enter at least one non-zero food-group dietary intake value before predicting.

Predict My Risk

Figure 13: Expected Input Step 4: Dietary Intake and Auto-Computed Nutrient Totals

The dietary section will allow the user to provide relevant food or nutrient intake values. Where applicable, the application will automatically compute derived totals to minimize input burden and reduce encoding inconsistencies.

5.3 Risk Assessment Output

After the user submits the required information, the system will process the input through the selected calibrated machine learning pipeline. The expected output will include a calibrated hypertension risk score displayed through a user-friendly visual

gauge. If a calibration method such as Venn-Abers is selected, the system may also display uncertainty bounds around the predicted risk estimate.

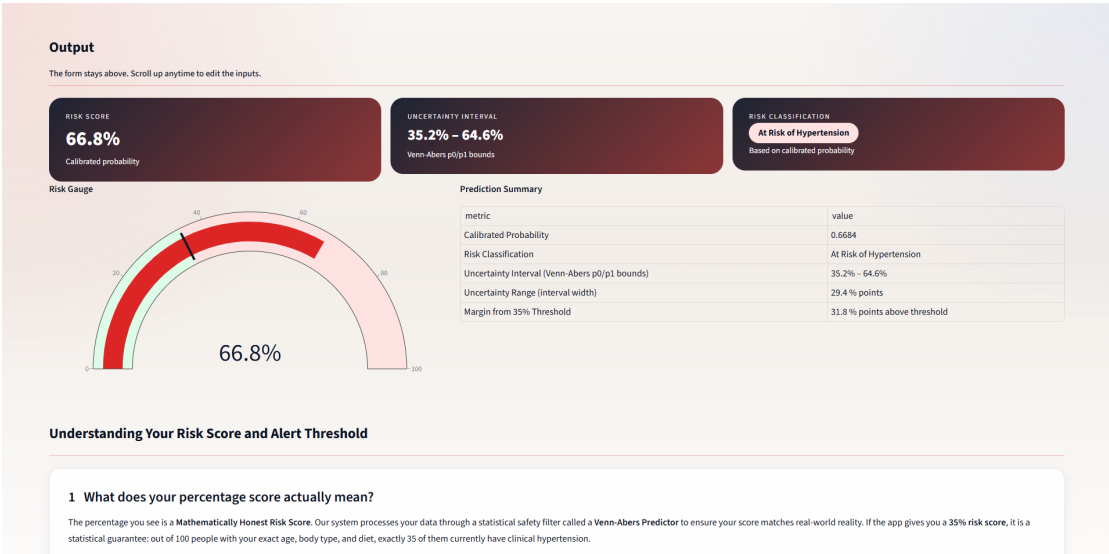


Figure 14: Expected Risk Assessment Output with Calibrated Risk Gauge and Uncertainty Display

5.4 Counterfactual Recommendation Output

The system is expected to provide actionable counterfactual suggestions based on the selected screening threshold. These counterfactuals will estimate the minimal feasible changes in actionable variables, such as anthropometric or lifestyle-related factors, that may reduce the modeled risk below the selected decision boundary.

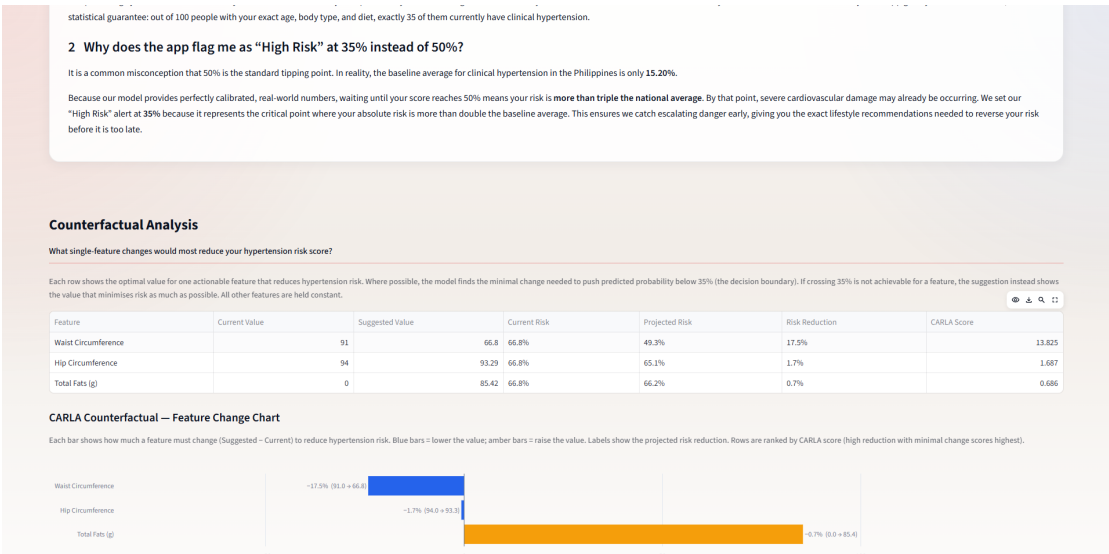


Figure 15: Expected Counterfactual Recommendation Output

5.5 Explainability Output

To improve transparency, the system will include explainability modules using SHAP and LIME. These outputs will help users or clinicians understand which features contributed most strongly to the predicted risk.

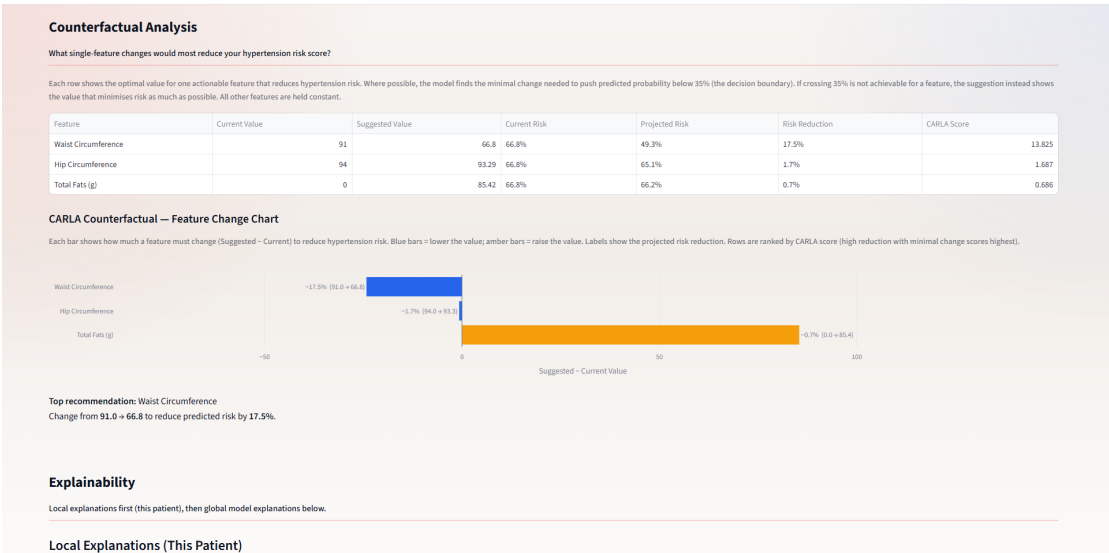


Figure 16: Expected Local SHAP Explanation Output

Local SHAP explanations will provide a patient-specific breakdown of how each feature contributes to increasing or decreasing the predicted hypertension risk.



Figure 17: Expected Local LIME Explanation Output

LIME explanations will provide an additional local interpretation of the model's decision in a more accessible visual format.

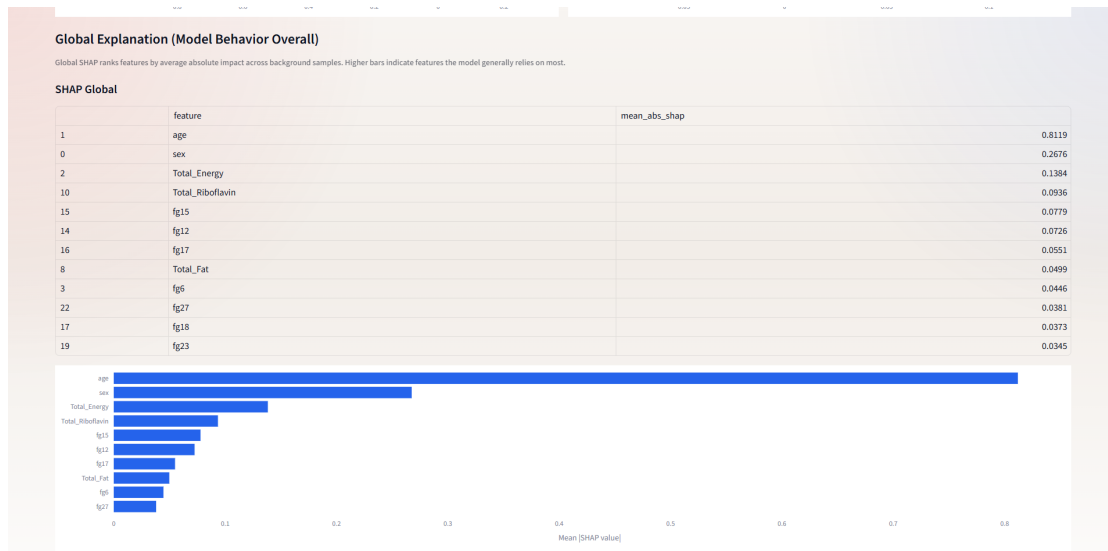


Figure 18: Expected Global SHAP Explanation Output

The Global SHAP output will provide population-level context by identifying the variables that generally contribute most strongly to the model’s predictions across the dataset.

5.6 Summary of Expected System Outputs

The completed prototype is expected to produce the following major outputs:

1. A progressive user input form for clinical, anthropometric, lifestyle, and dietary variables.
2. A calibrated hypertension risk score generated by the selected machine learning model.
3. An uncertainty interval when supported by the selected calibration method.
4. Local explanations using SHAP and LIME.
5. Global feature importance using SHAP.
6. Actionable Wachter-style counterfactual suggestions generated through bounded optimization.
7. A user-friendly web dashboard that presents these outputs in an organized decision-support interface.

6 Schedule of Activities

The proposed study will be conducted over a 24-week development period. The schedule is divided into five major phases: data acquisition and preprocessing, model development and optimization, calibration and explainability integration, system development, and final testing and documentation.

Table 5: Proposed Schedule of Activities

Phase	Activity	Duration	Description
Phase 1	Data Acquisition and Preprocessing	Weeks 1–4	Request and secure access to the 2015 DOST-FNRI NNS microdata, merge the Clinical and Health, Anthropometric, and Individual Dietary components, clean the dataset, handle missing values and outliers, and engineer the target and predictor variables.
Phase 2	Model Development and Optimization	Weeks 5–10	Implement the eight candidate machine learning algorithms, conduct randomized hyperparameter search, apply local refinement, and evaluate imbalance-handling configurations such as Baseline, Class Weights, SMOTE, SMOTENC, SMOTE + Class Weights, and SMOTENC + Class Weights.
Phase 3	Calibration and XAI Integration	Weeks 11–15	Apply probability calibration methods such as Platt Scaling, Isotonic Regression, and Venn-Abers predictors. Generate SHAP and LIME explanations and evaluate the models using discrimination and calibration metrics.
Phase 4	Web Application Development	Weeks 16–20	Develop the Streamlit-based prototype, integrate the selected calibrated model pipeline, implement the input interface, risk dashboard, explanation modules, uncertainty display, and counterfactual recommendation module.
Phase 5	Testing, Evaluation, and Writing	Weeks 21–24	Conduct system testing, evaluate final outputs, refine the interface, analyze results, prepare documentation, and complete the final manuscript.

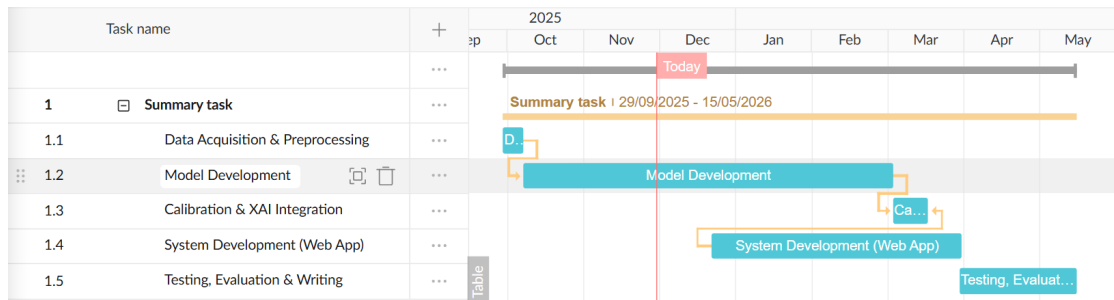


Figure 19: Proposed Gantt Chart of Project Development Timeline

The schedule is designed to ensure that the data processing, model development, calibration, explainability, and system implementation phases are completed sequentially while allowing sufficient time for testing and manuscript preparation. The modular division of work also allows the model training notebooks to be executed independently when computational resources or runtime limitations require separate execution.

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8 Appendix

8.1 Email Attachments - Data Collection

This section provides the documentation and approval emails regarding the acquisition of the 2015 National Nutrition Survey (NNS) datasets from the Department of Science and Technology - Food and Nutrition Research Institute (DOST-FNRI).

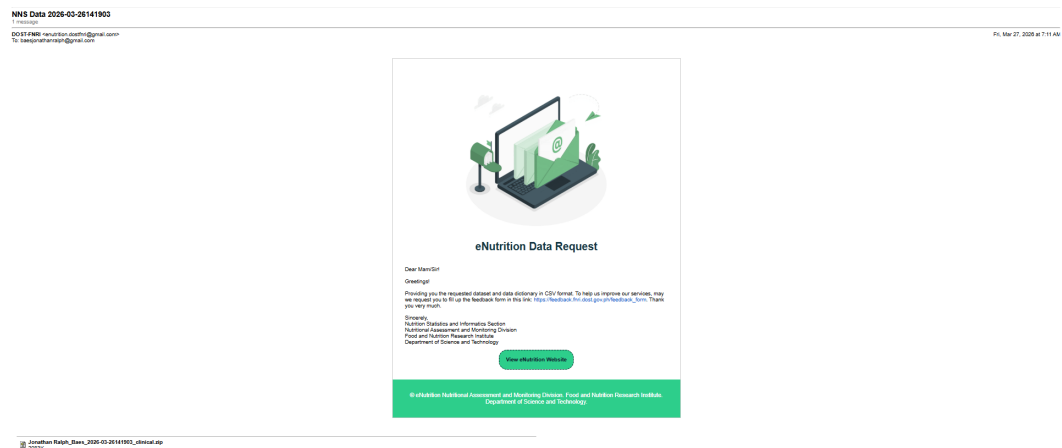


Figure 20: eNutrition Data Request Approval Email for Clinical and Health Component

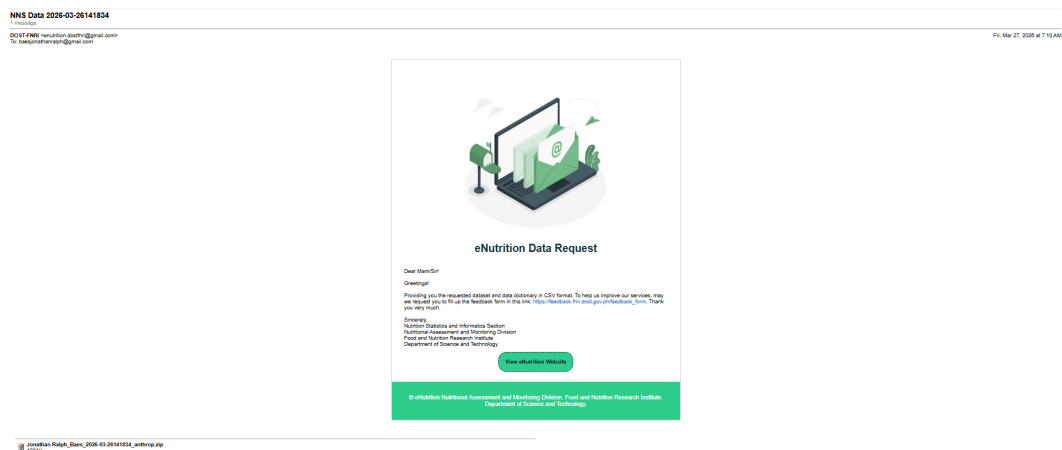


Figure 21: eNutrition Data Request Approval Email for Anthropometry Component



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